
Paracity marathon
6
7
WELCOME
TO Fabulous
LAS VEGAS
NEVADA
15
19
5
NO LAX ZONE
8
Utopia Resort
Put on toxic
a lid waste
10
4
3
16
Sign ups
2
11
9
13
21
22
1
18
17
12
CK
14
20
25
27
26
41
JUST TREAT IT
35
23
29
32
28
33
36
38
37
39
24
31
30
34
42
40

1.1 - Bradycardia DDx and SOAP

1) Slow “60” heart watch:
Bradycardia = **HR <60 bpm**

2) “Sign ups”:
sinus bradycardia (P precedes every QRS, upright P wave II, III, aVL, negative P wave aVR; distinguish from AV nodal block) due to inhibition or ↑parasympathetic on SA node → meds, organophosphate, vagal, metabolic, inf MI, ↑ICP

3) Beta bugle and calcium ice cream flowers:
Beta-blockers, Ca-channel blockers, and antiarrhythmics (e.g. amiodarone, lidocaine, digoxin) can cause sinus bradycardia/heart block)

4) “Glucagon” sweetener runner trampling beta flowers:
The treatment for beta blocker overdose includes IV fluids + atropine → **IV glucagon** → if no improvement = IV calcium salts, IV vasopressor (e.g. epinephrine), IV high-dose insulin and glucose, IV lipid emulsion therapy

5) Alpha flowers:
sympatholytic medications (e.g. **alpha-blockers** = clonidine, methyldopa) can cause sinus bradycardia/heart block

6) “Para-city”:
parasympathomimetic medications (e.g. acetylcholine, carbachol, acetylcholinesterase inhibitors) can cause sinus bradycardia/heart block

7) Relaxing “Utopia”:
Opioid intoxication can present with bradycardia → look for associated ↓level of consciousness, ↓breathing rate/tidal vol, miosis, ↓bowel sounds

8) “No lax zone”:
Treat opioid toxicity → manage ABCs and administer **IV naloxone** (opioid receptor antagonist)

9) Skunk spraying dumbbells:
Cholinergic toxicity (e.g. from organophosphate insecticide poisoning): can cause bradycardia with associated **DUMMBBELSS** symptoms (diarrhea, urination, miosis, muscle weakness, bronchospasm, emesis, lacrimation, salivation, and sweating)

10) “Put a lid on toxic waste” with Alice:
Organophosphate poisoning (cholinergic toxicity) is treated with immediate decontamination, **atropine** (block Ach receptors), and **pralidoxime** (reactivate Ach-esterase)

11) Athletic hare sleeping under “Vegas” sign:
increased vagal tone (carotid massage, valsalva, coughing/vomiting; physiologically normal in **athletes, sleeping**) can cause sinus bradycardia

12) Inferior heart lute getting dragged under “Vegas” sign :
inferior MI can present with sinus bradycardia (↑vagal tone, ↓perfusion of SA node)

13) Edematous pantaloons, dilated jug flowers, and angina anvil:
inferior MI classically presents with bradycardia, ischemic changes on ECG (II, III, aVF), ↑trops/CKMB, **chest pain**, and **right heart failure** (JVD, peripheral edema, clear lugs)

14) Elevated “ST” signs, troponin steak:
inferior MI classically presents with bradycardia, **ischemic changes on ECG** (II, III, aVF), ↑**trops/CKMB** , chest pain, and right heart failure (JVD, peripheral edema, clear lugs)

15) Shock wires next to “Vegas” sign:
Neurogenic shock (e.g. in setting of cervical/thoracic spinal trauma) → dysrupt sympathetic innervation of heart → bradycardia

16) Metabolic beverage table:
Metabolic causes of bradycardia = hypoxia, sepsis, hypothermia, hypoglycemia, hyperkalemia, anorexia

17) Undone thyroid bowtie:
Myxedema coma from profound hypothyroidism can cause bradycardia → look for associated hypothermia, hypotension, and altered mental status

18) T.S. Haberdasher:
test for ↑**TSH** in patient with bradycardia due to suspected hypothyroidism

19) Panda in pressurized head balloon struggling to breath while lifting steaming friend:
Cushing triad = **bradycardia, hypertension, and respiratory depression** in patient with central neurologic trauma/injury

20) Black heart hurdle:
AV nodal block due to inhibition or ↑parasympathetic on AV node (e.g. physiologic, vagal maneuvers, medications, metabolic); fibrosis/sclerosis of conduction system (idiopathic ~50% of cases); **ischemic/MI** (~40% of cases), myocarditis, cardiomyopathy, infiltrative disease

21) Amyloid armor and sarcoid soccer ball:
AV nodal block due to inhibition or ↑parasympathetic on AV node (e.g. physiologic, vagal maneuvers, medications, metabolic); fibrosis/sclerosis of conduction system (idiopathic ~50% of cases); ischemic/MI (~40% of cases); **myocarditis, cardiomyopathy, infiltrative disease**

22) Robin of ixodes with target and camping tent:
lyme disease **(Ixodes tick, wooded areas esp NE, targetoid rash)** → **lyme carditis**(**bradycardia, AV block** → palpitations, lightheadedness, syncope)

23(O) ECG fence at the starting line:
Confirm **sinus bradycardia** (P precedes every QRS, upright P wave II, III, aVL, negative P wave aVR) vs **AV nodal block** vs **junctional/ventricular escape rhythm (ECG/Bradycardia)**

24(S) Strained breath + sinking sloth clutching chest:
Bradycardia itself is typically **asymptomatic**; patients with comorbidities → **dyspnea, decreased exercise tolerance, lightheadedness, syncope, palpitations and/or chest pain→ (Chief Complaint/Bradycardia)**

25(S) ==O== Broken heart strings:
Rule out MI as cause of bradycardia (inferior MI classically presents with bradycardia, ischemic changes on ECG (II, III, aVF), ↑trops/CKMB, chest pain, right heart failure)] →**(HPI/Bradycardia)**

26) Black heart history:
Assess for cardiac etiology of bradycardia → history of CAD, MI, and heart failure **(PMHx/Bradycardia)**

27) Assessing the field of competitors:
Obtain a complete past medical history (bradycardia ddx= vagal, metabolic, infectious, myocarditis, cardiomyopathy, infiltrative disease, trauma-related) **(PMHx/Bradycardia)**

28(S) Med bag:
Assess for medication-related bradycardia (e.g Beta-blockers, CCBs, antiarrhythmics, sympatholytics, parasympathomimetics, etc.) **(Medications/Bradycardia)**

29(S) Hiking backpack:
Assess exposure to **industrial pesticides** (organophosphates) and **travel history**(wooded areas esp NE, lyme disease) **(Social Hx/Bradycardia)**

30(O) Slow heart watch:
Bradycardia = **HR <60 bpm (Vitals/Bradycardia)**

31(O) Hypotensive sloth:
Assess for **hemodynamic instability** (hypotension, chest discomfort, altered mental status, or shortness of breath) → immediate treatment! **(Vitals/Bradycardia)**

32(O) Irregular notes:
Auscultate/inspect ECG for **regular** (sinus bradycardia, 1st/3rd degree heart block, junctional/ventricular escape rhythm) vs **irregular** (2nd degree heart block) rhythm **(Physical exam/Bradycardia)**

33(O) “Slushi3” and stiff “4” chair:
Ascultate for new **S3 and S4** heart sounds → cardiac etiology (ischemia, cardiomyopathy) **(Physical Exam/Bradycardia)**

34(O) Failing heart balloon:
Assess for signs of **heart failure** (JVD, lower extremity edema, crackles) → may be exacerbated pre-existing HF, bradycardia-related new HF, sign of MI/ischemia **(Physical Exam/Bradycardia)**

35(P) Unstable lemming:
Unstable bradycardia (hypotension, chest discomfort, altered mental status, or shortness of breath) → IV atropine → refractory to therapy (consider dopamine, epinephrine) temporary pacemaker **(Plan (1)/Bradycardia)**

36(P) Alice catching unstable lemming:
Unstable bradycardia → **IV atropine** → refractory to therapy (consider dopamine, epinephrine) temporary pacemaker **(Plan (2)/Bradycardia)**

37(P) Temporary heart drummers:
Unstable bradycardia? → IV atropine → refractory to therapy (consider dopamine, epinephrine) **temporary pacemaker (Plan (3)/Bradycardia)**

38) Shirtless heart drummer:
Transcutaneous pacing = electrodes are placed on the patient's chest (temporary treatment for unstable bradycardia) **(Bradycardia)**

39) Right chest pocket of heart drummer:
Transvenous pacing = electrodes advanced through venous system to right ventricle (temporary treatment for unstable bradycardia) **(Bradycardia)**

40(P) Permanent heart drummer statue:
Unstable bradycardia? → IV atropine → refractory to therapy (consider dopamine, epinephrine) temporary pacemaker → **no underlying etiology identified Permanent pacemaker (Plan (4)/Bradycardia)**

41(P) Stable "Just treat it" cat:
Stable symptomatic bradycardia → r/o MI → assess for other underlying causes → check medications → permanent pacemaker if persistently symptomatic bradycardia, irreversible mobitz II AV nodal block, or irreversible 3rd degree AV nodal block **(Plan (4)/Bradycardia)**

42(P) "2II" and "3" place lemmings holding permanent heart drummer statues:
Stable symptomatic bradycardia? → r/o MI → assess for other underlying causes → check medications → **permanent pacemaker if persistently symptomatic bradycardia, irreversible mobitz II AV nodal block, or irreversible 3rd degree AV nodal block (Plan (5)/Bradycardia)**